Refutation of Natal-Al-saadi (arXiv:2610.16266)

CONFIRME

Refutation of NatalAl-saadi (arXiv:2610.16266) triangle bound for Conway's 99-graph

Status: CONFIRMED (passed the skeptic gate) the paper's experimental claim is false. Conway's 99-graph existence problem remains open.

Date: 2026-10-15

Short name: c

Statement

Natal & Al-saadi (arXiv:2610.16266v1, 12 Oct 2026) claim that every simple graph on 99 vertices with no diamond subgraph and no (K2,3) subgraph has at most 55 triangles (Conjecture 5 / Lasserre experiment). Since any ((99,14,1,2)) has exactly 231 triangles, they conclude that Conway's 99-graph does not exist.

We give explicit counterexamples with 66 and 120 triangles obeying both constraints (including connected graphs). Conjecture 5 is false; the paper does not resolve Conway's problem.

Background

Conway's 1000 problem asks whether ((99,14,1,2)) exists. The triangle count 231 is classical. As of 2026-10-15 the existence question remains open (ThakkarSeverini reductions; Labianco branch exclusions; ongoing SAT work). The NatalAl-saadi manuscript labels the key inequality a conjecture and states that a formal sum-of-squares certificate remains future work.

Paper (do not redistribute the PDF here): arXiv:2610.16266.

Forbidden-subgraph dictionary

- Diamond-free: no adjacent pair has 2 common neighbors.

- (K2,3)-free (

Necessary for (lambda=1), (mu=2), but not sufficient for strong regularity.

Counterexamples

Packing of ((9,4,1,2)) (66 triangles)

Let (L) be the unique ((9,4,1,2)) (33 lattice / Paley(9)). It has 6 triangles and is diamond-free and (K2,3)-free. The disjoint union of 11 copies of (L) has 99 vertices and

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Reproduce: python3 buildcounterexample.py

Hamming packing (120 triangles, connected) strongest

Let (H=H(4,3)=K3square K3square K3square K3) (81 vertices, 108 triangles, max common neighbors 1 on edges and 2 off edges). Add two copies of Paley(9) and two bridge edges. The resulting connected graph (C) on 99 vertices has 120 triangles and remains diamond-free and (K2,3)-free.

Reproduce (Skeptic B's independent code): python3 skeptics/B/verify.py

Extra: Fig. 2 (K2,3)-only claim

24 disjoint (K4) plus 3 isolates is (K2,3)-free with 96 triangles, so the caption claim that the bound is unchanged under only the (K2,3) constraint is also false (skeptics/B/k23only.py).

Why the nonexistence implication fails

An upper bound cannot sit below a triangle count achieved by an explicit graph. Since (T(99)ge 120>55), the reported SoS/Lasserre number cannot be a valid bound on the stated class. Without that bound, the paper only restates known facts (Lemma 1 and the 231-triangle count). Conway's problem stays open.

Skeptic verdicts

Skeptic A REJECT bound; REJECT proves nonexistent

Independent rebuild: 11 lattice ((9,4,1,2)) 66 triangles, constraints verified. See skeptics/A/VERDICT.md and skeptics/A/verifycounterexample.py.

Skeptic B REJECT bound; REJECT proves nonexistent

Independent rebuild: connected Hamming+Paley graph (C) 120 triangles; also confirms the 66-triangle packing. See skeptics/B/VERDICT.md and skeptics/B/verify.py.

Both skeptics started fresh, wrote their own code, and did not share workspaces with each other or with the hunt leads first draft directory during verification.

Reproduction

bash

python3 build

Honest caveats

- This refutes the triangle-bound nonexistence argument, not the existence of ((99,14,1,2)) itself.

- We did not r

Suggested author note (draft not sent)

> We enjoyed your preprint arXiv:2610.16266. Conjecture 5 appears to be false: the connected graph formed from the Hamming graph H(4,3) plus two copies of Paley(9), joined by two bridges, is diamond-free and K2,3-free on 99 vertices with 120 triangles (also: 11Paley(9) gives 66). So the reported T(99)55 bound cannot hold, and the nonexistence conclusion for (99,14,1,2) does not follow. Happy to share code: https://github.com/theesfeld/unsolved-hunter/tree/main/solves/conway99-sos-refutation

References

- J. Natal, O. Al-saadi, arXiv:2610.16266, 2026.

- A. E. Brouw